

National Institute of Fashion Technology

राष्ट्रीय फैशन प्रौद्योगिकी संस्थान

NIFT's VisioNxt प्रयोगशाला

WEAVING AI AND EI INTO THE FABRIC OF INDIA'S FASHION FUTURE

FASHION FUTURES: AI AND EI FOR TRENDS AND FORECASTING

In an era where fashion innovation meets style, designers and retailers require advanced tools to understand and design according to customer preferences ahead of time. Trend forecasting, driven by data analysis and cultural insights, has become essential for brands to thrive in a rapidly changing environment, especially in a pluralistic and culturally diverse nation as India. The global AI in fashion market is projected to reach USD 49.07 billion by 2033, growing at a CAGR of 41% (Precedence Research, 2024). AI and big data integration offer substantial opportunities for innovation, efficiency, and new skill sets, especially in India, where native AI-based forecasting systems are emerging.

NIFT'S VISIONARY INITIATIVE

The National Institute of Fashion Technology (NIFT), India's premier fashion education institute, is leading this transformation. The establishment of the VisioNxt प्रयोगशाला, a Trend Insight and Forecasting Lab, sanctioned by the Ministry of Textiles, marks a significant leap. This lab has developed India's first indigenous AI-EI-enabled fashion trend forecast system called "DeepVision" and published the first ever fashion forecast for India.

THE ROLE OF AI AND EI IN FASHION FORECASTING

AI (Artificial Intelligence) simulates human intelligence to analyze vast data, recognize patterns, and provide insights. EI (Emotional Intelligence) refers to understanding and managing emotions, enhancing decision-making and creativity. The VisioNxt project merges these technologies to interpret India's aesthetic and socio-cultural diversity, mapping trends in style, color, and regional accents.

LAUNCH OF PARIDHI 24X25

VisioNxt launched its first AI-EI-enabled trend book, Paridhi 24x25, on September 5, 2024, by Minister of Textiles, Shri Giriraj Singh. Post-launch, the VisioNxt e trend book in the website received over 23,000 visitors, and was downloaded by more than 2,000 industry and academic professionals. This initiative boosts India's global visibility in fashion forecasting, creating a unique Indian fashion identity and taxonomy for machine learning applications.

INDIGENOUS AI DEEP LEARNING MODEL

VisioNxt प्रयोगशाला incorporates AI deep learning models trained on over 2,00,000 image datasets of Indian wear. The project's research output includes three internationally published papers on AI, computer vision, and fashion image recognition. Over 800 NIFT students have gained new skills in trend spotting through the TrendOrb program, structured around six pillars: Scan, Capture, Map, Cluster, Analyze, and Present. This rigorous training equips students to decipher complex visual data and identify emerging trends.

COLLABORATIVE RESEARCH AND ENGAGEMENT

The lab engages AI engineers, fashion experts, and more than 20 academicians to strengthen trend intelligence. Students, researchers, and designers contribute to blogs, articles, trend snippets, and creative visuals, enriching India's trend ecosystem. VisioNxt's social media platform fosters engagement with trend updates and discussions. Additionally, the lab launched NXTMAG, India's first trend e-zine featuring articles on emerging mindsets, aesthetics, beauty, and fashion.

FUTURE IMPACT OF AI IN FASHION

Generative AI is expected to add $150-275 billion to the global fashion industry over the next 3-5 years (McKinsey & Company, 2023). With 62% of fashion leaders already using AI tools, VisioNxt is poised to make a significant impact. The lab not only provides trend insights but also equips students with industry-centric skills, ensuring India's fashion industry remains at the forefront of innovation and forecasting.

CONCLUSION:

VisioNxt प्रयोगशाला is revolutionizing fashion forecasting in India by blending AI and EI to decode trends and cultivate new skill sets. This initiative positions India prominently on the global fashion map while empowering students and professionals with cutting-edge tools. As AI's influence in fashion grows, VisioNxt ensures India leads the way in trend innovation and cultural expression.